

Digital System Design Mid Exam Syllabus

1. Introduction: Digital and Analog System.

2. Computer Organization

3. Number System : • Definition
• Conversion
• Boolean Algebra

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Prefixivity

→ • Prefixation
• Gray code

4. Boolean Algebra : • De-Morgan law

• সত্যতা প্রকাশ law

• logic simplification

• Universality of NAND Gate

• all Gate

5. Half Adder

6. Full Adder

7. Encoder

8. Decoder

DSD Mid Exam Suggestion by - Lopa Ma'am

Chapter: 1

1. What is digital system and analogue system.
The difference among them.
2. What are the advantage of digital system.
Over analog system,
3. What are the digital number system describe
~~strictly~~ briefly.
4. Describe the organization of digital computer
with figure.

* Short Question:

- BCD code
- Excess-3 code
- ASCII - Code
- Byte, Nibble and Word
- Unicode

Chapter :- 2

1. Number System
2. Describe the difference between BCD and Binary code.
3. Discuss how parity method is used for error detection.
4. Describe a four bit parity generator circuit with diagram.
5. 2's Complement and subtraction and addition.

Chapter :- 3

1. What are the basic Boolean expression. Explain that with Truth table. given the simple representation of gate.
2. Proof of logic expression using boolean theorem. ***

3. Which gate are known as Universal gate Describe them.
4. State and Prove De-Morgan theorem.
5. Discuss the alternative representations of logic gate.

Chapter: 4

1. Discuss the true forms representation of logic circuit.
2. The Problem of simplify logic expression and circuit using K-map and without K-map
3. Describe XOR and XNOR with circuit diagram and truth table
4. Parity generator
5. Problem of K-Map. ★★★
6. Previous year Question solve.